

A Factor-Two Lower Bound for Complementing Two-Way Deterministic Finite Automata

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Abstract

For every $n \geq 22$, we construct a language over a fixed three-letter alphabet whose two-way deterministic state complexity is n and whose complement has state complexity $2n$. Acceptance is at the left endmarker, and infinite computations reject. The lower bound holds for arbitrary deterministic recognizers. The proof uses the group of transformations induced by the witness to constrain the behavior of a smaller recognizer, then derives a contradiction from a permutation query problem. A matching halting complement follows from a $2n$ -state construction for sources whose destination states determine the direction of each transition. Splitting states by incoming direction gives a $2n + 2b$ upper bound for general sources, where b counts states entered in both directions. In the acceptance-anywhere model of Geffert, Mereghetti, and Pighizzini, the same family has exact complexities N and $2N - 1$ for every $N \geq 23$.

1. Introduction

Complementation is a basic operation on formal languages. If $L \subseteq \Sigma^*$, then the complement of L , relative to the fixed alphabet Σ , is $\bar{L} = \Sigma^* \setminus L$. Given a machine for L , we ask how large a machine of the same type for $\bar{L}$ must be, when such a machine exists.

For a deterministic machine that halts on every input, complementation is obtained by exchanging accepting and nonaccepting outcomes. The situation changes for recognizers that may have infinite computations. A word can then fail to be accepted either because the computation halts without accepting or because it never halts. A complementing construction must account for both possibilities. Sipser's method provides a way of converting nonhalting behavior into halting behavior in space-bounded machines [17].

For complete one-way deterministic finite automata, complementation is immediate: the machine reads the input once, always halts, and so swapping final and nonfinal states recognizes the complement. Two-way deterministic finite automata (2DFA) still recognize only regular languages [13, 16], but their head can move in both directions. A 2DFA may therefore reject by looping rather than by halting, so complementation cannot in general be obtained simply by swapping accepting and nonaccepting states.

When Sipser's method is specialized to two-way finite automata, it gives a quadratic $O(n^2)$ upper bound for complementing an n -state 2DFA. Geffert, Mereghetti, and Pighizzini (GMP) improved this to $4n$ states, independently of the input alphabet [2]. They asked whether complementation can require any increase at all: namely, whether there exists an n -state 2DFA whose complement cannot be recognized by an n -state 2DFA. This is the fifth open problem in their concluding section [2, Section 6, p. 1186, problem (e)].

We answer this question affirmatively under the acceptance convention fixed below. Here $sc(L)$ denotes the *state complexity* of L : the minimum number of states of a 2DFA recognizing L with acceptance at the left endmarker and infinite computations counted as rejection. For each $n \geq 22$, we construct an

n -state 2DFA over the fixed alphabet $\Sigma = \{A, B, C\}$ whose language K_n satisfies

$$\text{sc}(K_n) = n, \quad \text{sc}(\Sigma^* \setminus K_n) = 2n.$$

Both lower bounds apply to arbitrary deterministic recognizers in this model. The same family also answers GMP’s question in their original model, where a final state accepts at any tape position: its exact source and complement complexities are N and $2N - 1$, for every $N \geq 23$. Section 8 proves this comparison. The leading factor is two in both conventions.

The witness divides its states into two sets, or *banks*, one for forward scans and one for backward scans. Each input word induces a transformation on each bank. A forward scan followed by a backward scan tests an acceptance condition. If the first test fails, the machine makes a second round from a different starting state. If this test also fails, the same round repeats forever. Thus every rejecting computation of the witness is infinite.

We then consider an arbitrary automaton for $\overline{K_n}$. Its behavior on a nonempty input interval is summarized by the state and side of the first exit for every possible entry. We call this record an *interval diagram*. Equal diagrams can be substituted in any fixed context without changing acceptance. The witness supplies contexts that distinguish independent permutations on the two banks. These distinctions force a group of target diagrams to map onto a product of two alternating groups. This group acts on the states in which runs cross the interval, with the two exit directions counted separately. We call these signed exits *through channels*. Determinism bounds their number by the number of target states.

If the complementing machine had fewer than $2n$ states, we show that one alternating factor would act naturally on just one orbit of channels while fixing all other channels. On this subgroup, each traversal through the moving orbit amounts to a query to a permutation or its inverse. An answer leaves the machine in a fixed state at a fixed boundary, so the next action depends only on that answer and cannot also depend on which query produced it. We prove that such a controller cannot perform the test required by complementation. This forces at least $2n$ states. The same group argument also proves that the witness language itself requires n states.

A matching upper bound follows from reverse traversal of the configuration graph. In the witness, every transition entering a given state moves in the same direction. For sources with this property, called *direction-determinate*, the traversal uses $2n$ states. We give its full transition table for the present model. For an arbitrary source, splitting the b states entered in both directions gives a halting complement with at most $2n+2b$ states. Thus a lower bound with a leading constant greater than two would require linearly many source states entered in both directions.

Related work. State costs of simulations between one-way and two-way automata were studied by Sakoda and Sipser [14]; Pighizzini [11] surveys subsequent work. Jirásková and Okhotin [4] study operations on two-way automata, and Kunc and Okhotin [6] treat unary alphabets. Kapoutsis [5] studies one-way liveness on inputs of length two and three. Our alphabet has three letters, with no restriction on input length. The reverse traversal for direction-determinate automata belongs to the graph-walking constructions of Kunc and Okhotin [7]; its $2n$ state bound is restated by Martynova and Okhotin [8, Lemma 8].

For background on semigroups and permutation groups, see Pin [12], Howie [3], and Dixon and Mortimer [1]. Our small-index argument uses Shareshian and Woodroffe [15, Lemma 4.2]. Related descriptions of interval behavior occur in OpenAI’s preprints on deterministic simulation of one-way liveness [10] and nondeterministic complementation [9]. Those papers concern different state-complexity questions. The proof here gives the interval-diagram and group arguments needed for this lower bound, apart from the cited standard group-theoretic facts. No unproved conjecture is assumed.

2. Preliminaries and main result

2.1. Automata and acceptance

A 2DFA is a five-tuple $A = (Q, \Sigma, \delta, s, f)$. Here Q is a nonempty finite set of states, Σ is a finite ordinary alphabet, $s, f \in Q$, and

$$\delta : Q \times (\Sigma \sqcup \{L, R\}) \rightarrow Q \times \{-1, +1\}$$

is a partial function. The endmarkers L, R are not in Σ . Defined transitions on L move right, and defined transitions on R move left. On a word $w = x_1 \cdots x_t$, the input occupies positions $1, \dots, t$, with endmarkers at 0 and $t + 1$. A configuration is a pair (q, i) of a state and a tape position.

The computation begins in $(s, 0)$ and accepts exactly upon reaching $(f, 0)$. Acceptance takes priority over any outgoing transition there. An undefined transition at any other configuration halts and rejects; an infinite computation also rejects. In particular, state f is not accepting at an ordinary cell or at the right endmarker. There are no stationary moves or uncounted halting controls. Every member of Q , including s and f , contributes to the state count.

2.2. State complexity

Write $\text{sc}(L)$ for the minimum number of states of a 2DFA recognizing L in this model. Fix $\Sigma = \{A, B, C\}$, write $\bar{L} = \Sigma^* \setminus L$, and, for $n \geq 1$, set

$$C_3(n) = \max\{\text{sc}(\overline{L(A)}) : A \text{ is over } \Sigma, |Q_A| \leq n\}. \quad (1)$$

The maximum exists: there are finitely many transition tables up to state renaming, and the recognized languages are regular. Their complements can also be realized in this model by a rightward finite-state scan followed, upon success, by an accepting leftward sweep. The bound in (1) is on the number of states of the given source. We will also prove that no smaller source recognizes our witness language.

Theorem 2.1 (Exact factor-two witness). *For every $n \geq 22$, there is an explicitly specified n -state 2DFA over Σ whose language K_n satisfies*

$$\text{sc}(K_n) = n, \quad \text{sc}(\overline{K_n}) = 2n.$$

Consequently $C_3(n) \geq 2n$ for every $n \geq 22$. Both lower bounds range over arbitrary deterministic recognizers in the stated model, including recognizers with infinite rejecting computations.

Put $k = \lfloor n/2 \rfloor$ and $\ell = \lceil n/2 \rceil$. Then $k \geq 11$, $\ell \in \{k, k + 1\}$, and $n = k + \ell$. We construct $K_n = L_{k, \ell}$ below. Functions and permutations are composed from right to left. For a positive integer t , write $[t] = \{1, \dots, t\}$, and let $\mathfrak{S}_t$ and $\mathfrak{A}_t$ be the symmetric and alternating groups on $[t]$.

3. The ternary witness

3.1. The source five-tuple and its language

Fix integers $k \geq 11$ and $\ell \in \{k, k + 1\}$. For every $t \geq 11$, define

$$c_t = \begin{cases} (1 \ 2 \ \cdots \ t), & t \text{ odd,} \\ (1 \ 2 \ \cdots \ t-1), & t \text{ even,} \end{cases} \quad d_t = \begin{cases} (1 \ 2), & t \text{ odd,} \\ (t-1 \ t), & t \text{ even,} \end{cases}$$

and let e_t fix every point except 1, which it sends to 2. The order r_t of c_t is odd. Moreover, c_t, d_t generate $\mathfrak{S}_t$: in odd degree these are a full cycle and an adjacent transposition; in even degree the conjugates of d_t by powers of c_t are all transpositions $(i \ t)$, $1 \leq i < t$.

Use the fixed ordinary alphabet $\Sigma = \{A, B, C\}$. Associate to each letter two transformations as follows:

x	$\varphi_x : [k] \rightarrow [k]$	$\psi_x : [\ell] \rightarrow [\ell]$
A	c_k	d_ℓ
B	d_k	c_ℓ^{-1}
C	e_k	e_ℓ

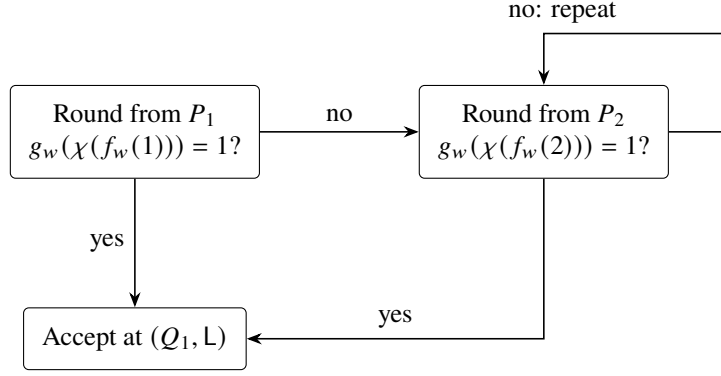


Figure 1: Each round consists of a forward scan in the P -states and a backward scan in the Q -states. All later rounds begin in P_2 . The boxes describe phases of the displayed automaton, with no additional states.

The source is the following five-tuple, with exactly $k + \ell$ states:

$$Q = \{P_i : i \in [k]\} \sqcup \{Q_j : j \in [\ell]\}, \quad A_{k,\ell} = (Q, \Sigma, \delta, P_1, Q_1). \quad (2)$$

For every ordinary $x \in \Sigma$, its transitions are

$$\delta(P_i, x) = (P_{\varphi_x(i)}, +1), \quad \delta(Q_j, x) = (Q_{\psi_x(j)}, -1). \quad (3)$$

The defined marker rows are

row	transition
(P_1, L)	$(P_1, +1)$
(P_1, R)	$(Q_1, -1)$
$(P_i, R), i \neq 1$	$(Q_2, -1)$
$(Q_j, L), j \neq 1$	$(P_2, +1)$.

(4)

All remaining marker rows are undefined. As stipulated by the model, (Q_1, L) accepts immediately.

For $w = x_1 \cdots x_t$, put

$$f_w = \varphi_{x_t} \circ \cdots \circ \varphi_{x_1}, \quad g_w = \psi_{x_1} \circ \cdots \circ \psi_{x_t}, \quad f_\varepsilon = g_\varepsilon = \text{id}. \quad (5)$$

The different orders reflect the two scan directions. In particular, $f_{uv} = f_v \circ f_u$ and $g_{uv} = g_u \circ g_v$. Define $\chi : [k] \rightarrow \{1, 2\}$ by $\chi(1) = 1$ and $\chi(i) = 2$ for $i \neq 1$.

Lemma 3.1 (Witness language). *The displayed source recognizes*

$$L_{k,\ell} = \{w : g_w(\chi(f_w(1))) = 1 \text{ or } g_w(\chi(f_w(2))) = 1\}. \quad (6)$$

Every word outside this language produces an infinite run.

Proof. The initial forward scan starts in P_1 and reaches the right marker in $P_{f_w(1)}$. Its marker row starts the reverse scan in $Q_{\chi(f_w(1))}$. That scan reaches the left marker in $Q_{g_w(\chi(f_w(1)))}$, accepting exactly when its index is 1. Otherwise, its marker row starts a forward scan in P_2 . Every subsequent round tests $g_w(\chi(f_w(2))) = 1$ in exactly the same way. If this test fails, the configuration immediately after the next left-marker transition repeats. The empty word accepts on its first round, consistently with the formula. $\square$

Figure 1 summarizes the source computation.

3.2. Words for independent transformations

For a word over $\{A, B\}$, both transformations are permutations; write $h_w = g_w^{-1}$. Then

$$(f_{uv}, h_{uv}) = (f_v \circ f_u, h_v \circ h_u).$$

Thus A and B induce the pairs (c_k, d_ℓ) and (d_k, c_ℓ) in the coordinates (f_w, h_w) . Since r_k, r_ℓ are odd,

word	(f_w, h_w)
A^{r_k}	(id, d_ℓ)
A^{r_k+1}	(c_k, id)
B^{r_ℓ}	(d_k, id)
$B^{r_\ell+1}$	(id, c_ℓ)

Consequently these ordinary words generate $\mathfrak{S}_k \times \mathfrak{S}_\ell$. Equivalently, every independent pair of permutations (f_w, g_w) has a word representative. Representatives may always be made nonempty by inserting the identity word A^{2r_k} .

Lemma 3.2 (Paired singular generation). *For every pair of singular (nonbijective) transformations $f : [k] \rightarrow [k]$ and $g : [\ell] \rightarrow [\ell]$, there is an ordinary word w with $(f_w, g_w) = (f, g)$.*

Proof. First, independent permutation words conjugate C into a word representing any prescribed pair of elementary collapses. More explicitly, if u, v have transformation pairs (p, q) and (p^{-1}, q^{-1}) , then

$$(f_{uqv}, g_{uqv}) = (p^{-1}e_k p, qe_\ell q^{-1}).$$

The permutations p, q can be chosen independently.

Every singular transformation of $[t]$ is a permutation composed after a nonempty product of elementary collapses. Indeed, choose one representative of each nonempty fiber and collapse each other point onto that representative. A permutation then maps the representatives to their prescribed images. Every elementary collapse is idempotent, so repeating a factor consecutively lengthens the product without changing its value. We may therefore write, with the same positive integer r ,

$$f = \pi \circ a_r \circ \cdots \circ a_1, \quad g = \rho \circ b_r \circ \cdots \circ b_1,$$

where all a_i, b_i are elementary collapses. Choose an initial permutation word with pair (id, ρ) , then r words with pairs (a_i, b_{r+1-i}) in the order $i = 1, \dots, r$, and a final permutation word with pair (π, id) . Their concatenation has pair (f, g) : the first coordinate composes in reverse physical order, while the second composes in physical order, as required for the backward scan. $\square$

The lemma covers pairs in which both coordinates are singular. All contexts used below have this form. In particular, define the singular map η_t by

$$\eta_t(1) = 1, \quad \eta_t(j) = 2 \quad (j \neq 1).$$

It is idempotent and fixes both 1 and 2. When an inactive bank need only preserve these two distinguished points, η_t supplies a singular coordinate for the paired-generation lemma.

4. Interval diagrams and through channels

4.1. Composition and substitution

Fix an arbitrary target B with m states recognizing $L_{k,\ell}$ or $\overline{L_{k,\ell}}$. No structural assumption is made about its transition table. In particular, it may change direction on every ordinary symbol and may reject by looping.

For a nonempty ordinary word v , run B on just its interval of cells, stopping at the first step outside that interval. A left entry in state q means that the head is on its first cell in state q ; a right entry means that it is on its last cell in state q . There are four partial exit functions:

	exit left	exit right
enter left	L_v	F_v
enter right	B_v	R_v

(7)

The output is the state in which the head first exits. For a given entry there is at most one finite exit. If the run hits an undefined transition or remains inside forever, both applicable functions are undefined. These two failures may be identified because both reject and neither can interact further with the outside of the interval. There is no accepting configuration inside this interval: it contains only ordinary letters, whereas acceptance is confined to the actual left endmarker.

The complete interval diagram $D_v = (F_v, B_v, L_v, R_v)$ records these functions on all states and both entry sides. An entry or exit, also called a *port*, is specified by a state and a side. Ports on opposite sides are distinct even when they have the same state label.

Lemma 4.1 (Composition and substitution). *The diagram D_{uv} is determined by D_u, D_v . Equality of diagrams permits substitution between any fixed ordinary left and right contexts with the actual endmarkers and initial/final convention. There are finitely many diagrams for an m -state target.*

Proof. At the join of u and v , an exit to the right of u becomes a left entry into v with the same state, and a left exit of v becomes a right entry into u with the same state. Follow these instructions until an exterior exit or a failure occurs. If joins are crossed infinitely often, there is no exterior exit. Equivalently, the finite join-control graph eventually repeats a control, which also yields no exterior exit. This computes all four functions of D_{uv} from the two diagrams. It describes the actual run, so the composition is associative.

The same calculation with arbitrary fixed outside regions preserves the whole deterministic run at every visit to an interval boundary. A run terminating or accepting in an outside region is therefore preserved. A run failing inside, or making infinitely many boundary visits without accepting, remains rejecting. This proves substitution, including loops and the actual accepting left endmarker configuration. Finally, each of $2m$ entries has at most $2m$ exits or failure, so at most $(2m + 1)^{2m}$ possible complete exit tables exist. Some such tables need not be realizable, but this upper bound proves finiteness. $\square$

It is useful to specify the multiplication convention explicitly. Write $D_u \diamond D_v = D_{uv}$ for physical concatenation. For group arguments use instead

$$D_u D_v = D_v \diamond D_u = D_{vu}. \quad (8)$$

This reverses the order of concatenation so that $D_w \mapsto (f_w, g_w^{-1})$ respects ordinary composition of functions. The convention concerns only the notation for multiplication.

4.2. The quotient determined by the source transformations

For the fixed recognizer, let M consist of the diagrams of nonempty $\{A, B\}$ -words whose semantic pair (f_w, g_w^{-1}) belongs to $\mathfrak{A}_k \times \mathfrak{A}_\ell$. We call this the *semantic pair*: it records the source transformations, whereas D_w records the target behavior. The set M is a finite semigroup under (8). Write $G = \mathfrak{A}_k \times \mathfrak{A}_\ell$. Every element of G has an allowed word representative by the independent generation above.

Lemma 4.2 (Product semantic quotient). *The assignment*

$$\theta(D_w) = (f_w, g_w^{-1})$$

is a well-defined surjective homomorphism $M \twoheadrightarrow \mathfrak{A}_k \times \mathfrak{A}_\ell$.

Proof. We prove that distinct semantic pairs are separated by ordinary contexts. For a middle word with pair (π, σ) , put $g = \sigma^{-1}$. For contexts λ, ρ , the complete transformations are

$$f_{\lambda w \rho} = f_\rho \circ \pi \circ f_\lambda, \quad g_{\lambda w \rho} = g_\lambda \circ g \circ g_\rho.$$

Consider two pairs (π, σ) and (π', σ') , writing $g' = \sigma'^{-1}$.

If $g \neq g'$, choose c with $g(c) \neq g'(c)$. Take f_λ and f_ρ constantly 1, take g_ρ constantly c , and let g_λ map $g(c)$ to 1 and every other point to 2. Each context has two singular coordinates and therefore exists by Lemma 3.2. The complete forward transformation is constant 1, and the two reverse outcomes are respectively 1 and 2. Lemma 3.1 distinguishes the two words.

If $g = g'$ but $\pi \neq \pi'$, choose a with $\pi(a) \neq \pi'(a)$. Take f_λ constantly a , and let f_ρ map $\pi(a)$ to 1 and every other point to 2. Take $g_\rho = \eta_\ell$, and let g_λ map $g(1)$ to 1 and every other point to 2. These are again paired singular contexts. Since g is a permutation, $g(1) \neq g(2)$. For the first middle word both forward tests launch the reverse scan at 1 and accept; for the second both launch it at 2 and fail. The source answers, and hence the target answers, differ.

Equality of interval diagrams permits substitution in fixed ordinary contexts, so these separations prove well-definedness. Surjectivity follows from independent permutation generation. Finally,

$$\theta(D_u D_v) = \theta(D_{vu}) = (f_u \circ f_v, g_u^{-1} \circ g_v^{-1}) = \theta(D_u) \theta(D_v),$$

which proves the homomorphism law. $\square$

4.3. Extracting a group

For an idempotent e (that is, $e^2 = e$), the set $eMe = \{exe : x \in M\}$ is a monoid with identity e , called its *local monoid*. A *unit* is an element invertible in that monoid. The following elementary argument extracts a local monoid that is a group.

Lemma 4.3. *There is an idempotent $e \in M$ such that eMe is a group mapping onto G . Its identity and every element have nonempty actual word representatives.*

Proof. Every element of a finite semigroup has a positive idempotent power. For completeness, its powers eventually have a period d ; choose a multiple t of d large enough that t and $2t$ are in the periodic part. Then $x^{2t} = x^t$.

On idempotents use the partial order $p \leq e$ when $pe = ep = p$. Reflexivity and antisymmetry are immediate; transitivity follows by substitution in these two identities. Choose a minimal idempotent e . For $h \in eMe$, an idempotent positive power h^t satisfies $eh^t = h^t e = h^t$, so $h^t \leq e$ and hence $h^t = e$. Consequently every h is a unit with inverse h^{t-1} when $t > 1$, and inverse e when $t = 1$. Thus eMe is a group with identity e .

The only idempotent in G is its identity, so $\theta(e) = 1$. For any $x \in M$, the element exe has image $\theta(x)$. Thus eMe still maps onto G . Every element belongs to M and is therefore represented by a nonempty word. $\square$

Fix this idempotent diagram e . Write

$$U = \text{Im } F_e, \quad V = \text{Im } B_e. \quad (9)$$

These are signed sets: U consists of right-exit labels and V of left-exit labels. They are treated as disjoint even when they contain the same underlying state name.

4.4. The action on through channels

Lemma 4.4 (Unit channels). *Every unit g of eMe has the same return functions and through domains as e . There are unique permutations α_g of U and β_g of V such that*

$$L_g = L_e, \quad R_g = R_e, \quad F_g = \alpha_g \circ F_e, \quad B_g = \beta_g \circ B_e. \quad (10)$$

The action $g \mapsto (\alpha_g, \beta_g^{-1})$ is faithful and, under multiplication (8), is a homomorphism into $\mathfrak{S}(U) \times \mathfrak{S}(V)$.

Proof. Let h be the inverse of g . A finite left return of the first factor in a physical concatenation persists unchanged in the product; a right return of the last factor likewise persists. Since $g = e \diamond g \diamond e$, every left/right return of e is a return of g . Conversely, applying these observations to $g \diamond h = e$ and $h \diamond g = e$ gives every return of g as a return of e . Hence $L_g = L_e$ and $R_g = R_e$.

A failed exterior entry of the outer e in $e \diamond g \diamond e$ cannot become a successful entry of g . Conversely, if a defined left entry of e failed in g , it would fail in $g \diamond h = e$; the corresponding right-entry assertion uses $h \diamond g = e$. Thus the domains of successful left and right entries agree. After removing the equal return domains, the forward and backward through domains agree as well.

Every final right exit of $e \diamond g \diamond e$ uses a forward row of its last e , so $\text{Im } F_g \subseteq U$. If two forward entries have equal F_e outputs, their paths merge after the first factor in $e \diamond g = g$, giving equal F_g outputs. Conversely, equal F_g outputs merge in $g \diamond h = e$, giving equal F_e outputs. The functions F_g, F_e therefore have exactly the same fibers on their common domain, and hence the same number of image points. The image inclusion is consequently equality, and there is a unique permutation α_g satisfying (10). Reflection gives the statement for B_g and β_g .

To prove the multiplication law, we examine what happens at the join between two intervals. In $e \diamond e = e$, enter the join with a right-exit label $u \in U$ from the left factor. Such a label is produced by some forward entry of e . At the join, successive left returns of the right factor and right returns of the left factor use only L_e, R_e . This fixed chain must eventually reach a forward-domain entry $r_+(u)$ in the right factor. It cannot fail, loop, or exit to the exterior left, because the corresponding forward row of $e \diamond e$ equals that of e . Its ultimate output satisfies $F_e(r_+(u)) = u$.

For units g, h , the return rows and through domains are identical, so the join chain after a forward output u of g is exactly this same fixed chain. The output through h is now $\alpha_h(u)$. It follows that, in physical order, $\alpha_{g \diamond h} = \alpha_h \alpha_g$. Reflection gives $\beta_{g \diamond h} = \beta_g \beta_h$. Under (8) these laws become $\alpha_{gh} = \alpha_g \alpha_h$ and $\beta_{gh} = \beta_h \beta_g$, proving the stated homomorphism. Finally (10), together with the fixed return and failure rows, determines the whole diagram. Its representation by the pair of permutations is therefore faithful. In particular, $\alpha_e = \text{id}_U$ and $\beta_e = \text{id}_V$. $\square$

Lemma 4.5 (The number of through channels). *The signed through-image sets satisfy*

$$|U| + |V| \leq m. \quad (11)$$

Proof. Choose an actual nonempty word representing e and one through run for each distinct signed exit in $U \sqcup V$. Each chosen run visits every cell of that word. No two can contain the same state on the same cell, because the two deterministic suffix runs would then be identical and would give the same signed exit. Fix any one ordinary cell. The chosen runs consequently use distinct physical states there, giving (11). The argument uses one ordinary cell and therefore requires a nonempty word; an empty interval would formally have $2m$ through ports. $\square$

5. Small actions of a product of alternating groups

We next show why fewer than $2(k+\ell)$ channels force one alternating factor to act on just one natural orbit. The main step is Lemma 5.3. Its proof first limits the possible orbit sizes, then removes the actions on the other orbits by taking derived subgroups. The first two lemmas provide the required index bounds.

An action is *faithful* if only the identity fixes every point. A transitive permutation group is *imprimitive* if it preserves a partition into equal-sized blocks with sizes strictly between one and the degree; otherwise it is *primitive*. The derived series of a group K is defined by $K^{(0)} = K$ and $K^{(j+1)} = [K^{(j)}, K^{(j)}]$. A group is *solvable* if some term is trivial, and *perfect* if $K^{(1)} = K$. We use the standard facts that $\mathfrak{A}_d$ is nonabelian simple and perfect for $d \geq 5$, and that every subgroup of $\mathfrak{S}_r$ for $r \leq 4$ has derived length at most three; see [1].

We use the following theorem of Shalev and Woodroffe: for $d \geq 9$, every prime $p \leq d-3$ divides the index of every proper primitive subgroup of $\mathfrak{A}_d$. This is [15, Lemma 4.2, p. 230]. For odd

p , its proof uses Sylow conjugacy and Jordan's prime-cycle theorem. For $p = 2$, it uses the double-transposition part of Jordan's theorem, stated as [15, Theorem 4.1(2)]. Together with Bertrand's postulate, this implies the elementary small-index bounds below.

First recall that every proper subgroup of $\mathfrak{A}_d$, for $d \geq 5$, has index at least d . Indeed, its nontrivial coset action is faithful by simplicity. An index $r < d$ would therefore give $d!/2 \leq r! \leq (d-1)!$, a contradiction.

Lemma 5.1 (Subgroups of small index). *If $t \geq 11$ and $J < \mathfrak{A}_t$ satisfies $[\mathfrak{A}_t : J] < 4t + 2$, then J is the stabilizer of a point in the natural action on $[t]$.*

Proof. Suppose first that J is primitive. For $11 \leq t \leq 13$, the primitive-subgroup fact forces the distinct primes 2, 3, 5, 7 to divide its index. Their product 210 exceeds $4t+2$. For $t \geq 14$, put $M = \lfloor (t-3)/2 \rfloor \geq 5$. Bertrand's postulate supplies a prime $M < p < 2M \leq t-3$. This prime is at least 7, so the distinct primes 2, 3, 5, p divide the index. Consequently

$$[\mathfrak{A}_t : J] \geq 30p > 30M \geq 15(t-4) > 4t+2,$$

again a contradiction.

If J is transitive and imprimitive, it preserves a partition into blocks of a common size b , where $2 \leq b \leq t/2$ and b divides t . The group $\mathfrak{A}_t$ is transitive on such partitions: an odd transporter can be made even by a transposition within a block of the image partition. Thus the index of the partition stabilizer is the number of these partitions, and is a lower bound for $[\mathfrak{A}_t : J]$. For $b \geq 3$, choosing the other $b-1$ members of the block containing one specified point gives at least

$$\binom{t-1}{b-1} \geq \binom{t-1}{2} > 4t+2$$

partitions for $t \geq 12$. For $b = 2$ the number is $(t-1)!! \geq (t-1)(t-3) > 4t+2$, also for $t \geq 12$. There is no imprimitive case when $t = 11$, since 11 is prime.

Finally suppose that J is intransitive. Unless it has an orbit of size $t-1$, a union of its orbits has some size a with $2 \leq a \leq t-2$. Its setwise stabilizer has index $\binom{t}{a}$ in $\mathfrak{A}_t$, whence

$$[\mathfrak{A}_t : J] \geq \binom{t}{a} \geq \binom{t}{2} > 4t+2.$$

Here transitivity on a -subsets follows by correcting an odd transporter with a transposition inside its image subset. The remaining case fixes one point and has $J \leq \mathfrak{A}_{t-1}$. If this inclusion were proper, the minimum-index observation above would give $[\mathfrak{A}_t : J] \geq t(t-1) > 4t+2$. Therefore J is the full point stabilizer. $\square$

Lemma 5.2 (Small-index product subgroups). *Let $k \geq 11$, $\ell \in \{k, k+1\}$, and $G = \mathfrak{A}_k \times \mathfrak{A}_\ell$. Every proper subgroup $J < G$ with $[G : J] < 2(k+\ell)$ has one of the forms*

$$J = P \times \mathfrak{A}_\ell \quad \text{or} \quad J = \mathfrak{A}_k \times Q,$$

where P or Q , respectively, is a natural point stabilizer.

Proof. Let $P \leq \mathfrak{A}_k$ and $Q \leq \mathfrak{A}_\ell$ be the projections of J . If both are proper, then

$$[G : J] \geq [\mathfrak{A}_k : P][\mathfrak{A}_\ell : Q] \geq k\ell > 2(k+\ell),$$

which is impossible. Suppose, without loss of generality, that $Q = \mathfrak{A}_\ell$. The subgroup $J \cap (\{1\} \times \mathfrak{A}_\ell)$ is normal in $\{1\} \times \mathfrak{A}_\ell$, because the second projection of J is surjective. Simplicity makes this intersection trivial or the whole second factor. In the former case the first projection embeds J into P , and hence

$$[G : J] = \frac{|\mathfrak{A}_k||\mathfrak{A}_\ell|}{|J|} \geq |\mathfrak{A}_\ell| > 2(k+\ell),$$

a contradiction. Thus $J = P \times \mathfrak{A}_\ell$, with P proper since J is proper. Now $[\mathfrak{A}_k : P] < 2(k+\ell) \leq 4k+2$, so Lemma 5.1 applies. The other case is symmetric, using $2(k+\ell) \leq 4\ell+2$. $\square$

Lemma 5.3 (Isolation of one natural factor orbit). *Let $k \geq 11$, $\ell \in \{k, k+1\}$, and $G = \mathfrak{A}_k \times \mathfrak{A}_\ell$. Suppose a finite group H acts faithfully on a finite set X and admits a surjective homomorphism $\theta : H \twoheadrightarrow G$. If $|X| < 2(k + \ell)$, then H has a subgroup Γ with the following properties. For one $d \in \{k, \ell\}$, the restriction of θ is an isomorphism from Γ onto the corresponding factor $\mathfrak{A}_d$ (with the other factor equal to the identity), and Γ has exactly one nonfixed orbit on X . This orbit has size d and is the natural action of that same semantic factor. Every other point of X is fixed. If $X = U \sqcup V$ with both parts H -invariant, this orbit lies wholly in U or wholly in V .*

Proof. Choose an inclusion-minimal subgroup $\Lambda \leq H$ still surjecting onto G , and write $N = \ker(\theta|_\Lambda)$. The restricted action of Λ remains faithful. For a nonfixed point x , let $H_x = \Lambda_x$ and $J_x = \theta(H_x)$. Since H_x is proper, minimality makes J_x proper as well. The index formula gives

$$|\Lambda x| = [\Lambda : H_x] = [G : J_x] [N : N \cap H_x]. \quad (12)$$

Lemma 5.2 therefore assigns each nonfixed Λ -orbit to one of the two factors: its size is dr , where d is the degree of that factor and $r = [N : N \cap H_x] \geq 1$. Since $dr < 2(k + \ell) \leq 4d + 2$ and $d \geq 11$, necessarily $r \leq 4$.

More precisely, the N -orbits within Λx form d blocks, each of size r . Indeed, $NH_x = \theta^{-1}(J_x)$, so these blocks are indexed by $\Lambda/(NH_x)$. Their induced Λ -action is the coset action of G on G/J_x , which is the natural action of the assigned factor. The other factor of G fixes each of these blocks.

Each factor must have at least one orbit assigned to it. Otherwise, let K be the full preimage in Λ of that factor, with the other factor equal to the identity. On every nonfixed orbit, K would fix every N -block setwise. Its action would consequently embed into a product of symmetric groups on sets of size at most four; globally fixed points contribute trivial factors. Faithfulness would make K solvable, since each $\mathfrak{S}_r$ for $r \leq 4$ is solvable. This is impossible because K surjects onto the nonabelian simple alternating factor.

Label the two factors by 1, 2, with degrees $d_1 = k$ and $d_2 = \ell$, even when $k = \ell$. Let a_1, a_2 be the total numbers of points in the orbits assigned to them. We have $a_1 + a_2 \leq |X| < 2d_1 + 2d_2$. Thus for one degree d its assigned points number less than $2d$. This positive number is a sum of positive integer multiples of d , so it is exactly d . There is exactly one orbit assigned to this factor, and it has $r = 1$.

Let K again denote the full preimage of this factor in Λ . On every orbit assigned to the other factor, its image lies in a direct product of groups $\mathfrak{S}_r$ with $r \leq 4$: the blocks are fixed setwise, so there is no additional permutation of blocks. Such an image has derived length at most three: for $\mathfrak{S}_4$ the successive derived subgroups are $\mathfrak{A}_4$, the Klein four group, and the identity. Therefore $\Gamma = K^{(3)}$, the third derived subgroup of K , fixes all these other-factor orbits. It also fixes the globally fixed points.

Because $\mathfrak{A}_d$ is perfect, $\theta(\Gamma) = \mathfrak{A}_d$. On the unique orbit assigned to this factor the fibers have size one, so N acts trivially and the action is the natural action of the same semantic $\mathfrak{A}_d$. Explicitly, if J_x fixes point a in the selected factor i , label λx by $\theta_i(\lambda)(a)$. The stabilizer identity makes this labeling well defined and bijective, and $\gamma \in \Gamma$ acts on the labels by $\theta_i(\gamma)$. Any element of $\Gamma \cap N$ therefore fixes all of X ; faithfulness implies that it is the identity. Thus $\theta|_\Gamma$ is the required isomorphism. Finally any Γ -orbit is contained in an invariant part U or V , proving the signed-orbit assertion. $\square$

Corollary 5.4 (Minimum degree of the product action). *Under the group and degree hypotheses of Lemma 5.3, every faithful action of a finite group surjecting onto $\mathfrak{A}_k \times \mathfrak{A}_\ell$ has at least $k + \ell$ points. In particular, if a group of actual interval diagrams surjects onto this product, its faithful signed through-channel action satisfies $|U| + |V| \geq k + \ell$.*

Proof. An action with fewer than $k + \ell$ points is below the threshold $2(k + \ell)$. Apply the orbit-and-block analysis in the proof of Lemma 5.3: each factor must have an assigned orbit, and those two disjoint orbits have sizes at least k and ℓ . This contradicts the assumed size. The interval-diagram statement uses the faithful signed channel action of Lemma 4.4. $\square$

6. Permutation queries and the lower bounds

6.1. Fixed answer continuations

For an unknown $\pi \in \mathfrak{A}_d$, consider a *query controller*. Its initial instruction is a query index or a terminal outcome. Its answer continuation is a fixed function

$$t : [d] \longrightarrow [d] \sqcup \{\text{accept}, \text{reject}\}.$$

An instruction $a \in [d]$ obtains the raw answer $\pi(a)$ and then executes $t(\pi(a))$. Thus the answer retains neither the caller nor a separate copy of its query index. The terminal instructions accept and reject; an infinite sequence of queries rejects. Fixed helper computations can be folded into the instructions, with a fixed helper loop counted as rejection. The first answer uses the same function t as every later answer.

Lemma 6.1 (From one channel orbit to queries). *Suppose a subgroup of the interval-diagram group eMe , with semantic parameter $\pi \in \mathfrak{A}_d$, acts naturally on exactly one orbit of d signed through channels and fixes all other channels. The natural action is that of the same semantic permutation π . After fixing ordinary left and right contexts, acceptance on the corresponding middle-word representatives is decided by a query controller. Its raw operation is $a \mapsto \pi(a)$ when the moving orbit is forward, and $a \mapsto \pi^{-1}(a)$ when it is backward.*

Proof. Choose one nonempty word for each group element. The return functions, failure domains, and through domains are fixed by Lemma 4.4. A forward through row has the form $F_h(q) = \alpha_h(F_e(q))$, and a backward row has the form $B_h(q) = \beta_h(B_e(q))$. Outside the moving orbit these responses are constant. On that orbit, the forward response is $\pi(a)$ or the backward response is $\pi^{-1}(a)$, because the homomorphic channel action is (α_h, β_h^{-1}) .

Consider the left region consisting of the actual left endmarker and the fixed left context, and the right region consisting of the fixed right context and the actual right endmarker. Replace each visit to the middle interval by its complete exit response. Fold the two fixed regions, all return rows, and all constant through responses into a finite deterministic network. From a boundary configuration, follow this network until it accepts, rejects, enters a fixed cycle, or asks for a variable through response. In the last case its query index is fixed by that boundary configuration. Failure or a cycle gives the reject instruction.

A variable forward answer is a physical state at the first cell of the fixed right region. A variable backward answer is a physical state at the last cell of the fixed left region. Each natural-orbit label determines one such state, and all variable answers use the same side because the orbit lies wholly in U or wholly in V . The continuation therefore depends only on the answer label. This also covers empty ordinary contexts, since their regions still contain the corresponding endmarkers. Any fixed processing of a raw answer is included in its one continuation.

Start the same folding from the actual initial configuration. It either has a fixed terminal outcome or reaches a first query. The first raw answer arrives at exactly the same answer configuration as later occurrences of that label, with no initialization tag. Although middle representatives can have different lengths, positions inside each fixed region are identified relative to its boundary. The finite control has no clock or absolute-position register that could retain traversal length. These relative configurations and their transition rules are identical across representatives. The resulting initial instruction and fixed answer function give the stated controller, including all rejecting loops. $\square$

6.2. An inverse-query obstruction

Lemma 6.2 (Even completion). *For $d \geq 5$, every prescription of at most three distinct input–output pairs with distinct inputs and distinct outputs extends to an even permutation of $[d]$.*

Proof. Complete the prescribed injection arbitrarily to a permutation. If it is odd, interchange the images of two unassigned inputs. There are at least $d - 3 \geq 2$ such inputs; this changes the parity without changing a prescribed pair. $\square$

Lemma 6.3 (A limit on permutation queries). *For $d \geq 5$, a query controller cannot recognize*

$$\pi^{-1}(i) \in T \quad (13)$$

when $i \in [d]$ and $3 \leq |T| \leq d - 2$. Infinite runs count as rejection, as in the automaton model.

Proof. The predicate is nonconstant, so the initial fixed computation must query some index a . For each first answer $j \neq i$, the condition $\pi(a) = j$ is compatible with either truth value in (13): prescribe the preimage of i in $T \setminus \{a\}$ for an instance satisfying the predicate, or in $([d] \setminus T) \setminus \{a\}$ for one that does not. Both sets are nonempty. These are two-pair prescriptions and hence have even completions by Lemma 6.2.

Thus no answer $j \neq i$ may accept, reject, or follow only a fixed helper loop. It must query an index $f(j)$. If $a \notin T$, the first answer i occurs on a negative input, so its instruction may not accept either. There would then be no accepting instruction at any answer control. A finite accepting run would be impossible, although positive inputs exist. Therefore $a \in T$. Since the only answer that could accept is i , its instruction must accept. Otherwise none of the answer controls could terminate with acceptance.

If $f(j) = a$ for some $j \neq i$, choose any $c \in T \setminus \{a\}$ and an even permutation satisfying $\pi(a) = j, \pi(c) = i$. The predicate holds for this permutation, but the controller repeats query a and answer j forever. Thus $f(j) \neq a$. If $f(j) \notin T$, prescribe $\pi(a) = j$ and $\pi(f(j)) = i$. The computation accepts after the second query, although the prescribed preimage of i is outside T . Hence necessarily

$$f(j) \in T \setminus \{a\} \quad (j \neq i). \quad (14)$$

There are $d - 1$ arguments $j \neq i$ but fewer than $d - 1$ possible values in (14), so two distinct $j_1, j_2 \neq i$ have $f(j_1) = f(j_2) = b$. Choose $c \in T \setminus \{a, b\}$, possible because $|T| \geq 3$. Lemma 6.2 supplies an even permutation with

$$\pi(a) = j_1, \quad \pi(b) = j_2, \quad \pi(c) = i.$$

It satisfies (13), but its run asks a , then b , then b forever, and hence rejects. This is the required contradiction. $\square$

6.3. The complement lower bound

Theorem 6.4. *For $k \geq 11$ and $\ell \in \{k, k + 1\}$, every 2DFA recognizing $\overline{L_{k,\ell}}$ has at least $2(k + \ell)$ states.*

Proof. Suppose a target for the complement has $m < 2(k + \ell)$ states. Lemmas 4.2, 4.3, 4.4, and 4.5 give a faithful group of signed channels mapping onto $\mathfrak{A}_k \times \mathfrak{A}_\ell$, with at most m channels. Lemma 5.3 supplies a subgroup mapping isomorphically onto one semantic factor, with exactly one natural moving orbit for that factor, and fixing all other channels. Its other semantic coordinate is the identity. Every element has an actual nonempty word representative.

We now choose contexts that turn acceptance into the query test of Lemma 6.3. Each transformation used in these contexts is singular, so the contexts exist by Lemma 3.2. In an inactive coordinate use the rank-two idempotent η_d that fixes 1, 2 and sends every other point to 2. It need not act as identity on all points. For each active coordinate of degree d , a selector with $a(1) = x, a(2) = y$ can be chosen to send every index other than 1 to y . An output test for $S \subseteq [d]$ sends S to 1 and its complement to 2. Thus each active map also has rank at most two. All these maps are singular because $d \geq 11$.

The subgroup maps onto the forward factor. The semantic middle pair is (π, id) , with $\pi \in \mathfrak{A}_k$. Choose left and right forward transformations a, b with $a(1) = x, a(2) = y$, and $b^{-1}(1) = S$. Both backward context transformations are η_ℓ . The entire backward map fixes 1, 2, so the source answer is

$$\pi(x) \in S \quad \text{or} \quad \pi(y) \in S. \quad (15)$$

If the moving channel orbit is forward, take $x \neq y$ and $S = \{i\}$. The complement predicate is $\pi^{-1}(i) \in [k] \setminus \{x, y\}$, forbidden by Lemmas 6.1 and 6.3. If the orbit is backward, set $x = y = i$ and $|S| = 2$. Put $\sigma = \pi^{-1}$. The raw query is σ , and the complement predicate is $\sigma^{-1}(i) \in [k] \setminus S$, again forbidden.

The subgroup maps onto the backward factor. Write the semantic middle pair as (id, σ) , so the actual backward map is $g = \sigma^{-1}$. Give both forward contexts transformation η_k . The entire forward map fixes 1, 2. Put a selector a , with $a(1) = x, a(2) = y$, in the *right* backward context, and an output test b , with $b^{-1}(1) = S$, in the *left* backward context. The source answer is

$$g(x) \in S \quad \text{or} \quad g(y) \in S. \quad (16)$$

If the moving orbit is forward, its raw query is σ . Set $x = y = i$ and $|S| = 2$; the complement is $\sigma^{-1}(i) \in [\ell] \setminus S$. If the orbit is backward, its raw query is $g = \sigma^{-1}$. Set $x \neq y$ and $S = \{i\}$; the complement is $g^{-1}(i) \in [\ell] \setminus \{x, y\}$. Both predicates violate the same obstruction. This covers every possible isolated factor and orientation. Therefore $m \geq 2(k + \ell)$. $\square$

6.4. Exact source complexity

Corollary 6.5. *For $k \geq 11$ and $\ell \in \{k, k + 1\}$, $\text{sc}(L_{k,\ell}) = k + \ell$.*

Proof. Let an arbitrary m -state automaton recognize $L_{k,\ell}$. The semantic separation in Lemma 4.2 applies to this recognizer as well as to a complementing target. Its interval diagram semigroup therefore maps onto $\mathfrak{A}_k \times \mathfrak{A}_\ell$. Lemmas 4.3 and 4.4 give a group with a faithful action on the signed channels $U \sqcup V$. Corollary 5.4 gives $|U| + |V| \geq k + \ell$, while Lemma 4.5 gives $|U| + |V| \leq m$. Hence $m \geq k + \ell$. The five-tuple in Section 3 attains this bound. $\square$

7. Halting complementation and incoming directions

7.1. Direction-determinate sources

We give the reverse-tree traversal explicitly in the present model. The method goes back to Sipser [17]; its use in complementation appears in GMP [2]. The two-state-per-source-state traversal for direction-determinate graph-walking automata is given by Kunc and Okhotin [7] and restated in [8, Lemma 8]. We specify the initial state and accepting configuration as part of the transition table, so neither requires an additional state.

Call a source automaton *direction-determinate* if there is a map $d : Q \rightarrow \{-1, +1\}$ such that every defined transition, including every endmarker transition, has the form

$$\delta(p, a) = (q, d(q)).$$

Thus the state entered by a transition determines its direction. The two banks of our witness have this property: every P_i is entered moving right, and every Q_j is entered moving left.

Theorem 7.1. *Every direction-determinate n -state source has a halting complement with at most $2n$ states in the exact model of this paper.*

Proof. Write the source as $(Q, \Sigma, \delta, s, f)$. If $s = f$, its language is Σ^* ; a two-state automaton with a nonfinal initial state and no transitions recognizes its complement. If $s \neq f$ and $d(f) = +1$, the configuration (f, L) is unreachable, and a one-state automaton whose initial state is final recognizes the complement. Both constructions halt and use at most $2n$ states. Hence assume $s \neq f$ and $d(f) = -1$.

Let δ^* be δ with the row (f, L) deleted. Acceptance takes priority over this row, so its deletion does not change the source language. Fix a total order $<$ on Q . The complement has state set

$$Q' = \{B_q : q \in Q\} \cup \{F_q : q \in Q\}, \quad A^c = (Q', \Sigma, \delta', B_f, F_f).$$

These are all the states of the complement. Their roles are as follows. A configuration (B_q, j) enumerates the predecessors of the source configuration $(q, j + d(q))$; all those predecessors lie at cell j . A configuration (F_p, j) means that the reverse subtree rooted at (p, j) has been traversed, and the construction should advance to the next sibling or return to the parent.

Here is a complete finite prescription for the transition table. For each ordinary symbol or endmarker a , put

$$P(q, a) = \{p \in Q : \delta^*(p, a) = (q, d(q))\},$$

listed in increasing order. Given any increasing sublist C of Q , define $\text{select}(C, a)$ by inspecting its candidates in order:

1. If $a = L$ and the current candidate p is s , return reject.
2. If $a = L$ and $d(p) = +1$, or if $a = R$ and $d(p) = -1$, skip candidate p . Such a source configuration has no predecessor, since its predecessor cell would be outside the tape.
3. Otherwise return the transition $(B_p, -d(p))$.

If the list is exhausted, return exhausted. The selection is made when the transition table is constructed, not during the run. It therefore uses no extra states or stationary moves. In particular, the initial-configuration test precedes the boundary-leaf test.

Set the initialization row to

$$\delta'(B_f, L) = (B_f, +1).$$

Apart from this exceptional row, leave (B_q, L) undefined when $d(q) = -1$, and leave (B_q, R) undefined when $d(q) = +1$. For every other backward row (B_q, a) , evaluate $\text{select}(P(q, a), a)$. Use its returned transition; reject means that the row is undefined, and exhausted gives the transition $(F_q, d(q))$.

For a forward row (F_p, a) , the accepting configuration (F_f, L) needs no transition. If $\delta^*(p, a)$ is undefined, leave the row undefined. Otherwise write $\delta^*(p, a) = (q, d(q))$ and evaluate

$$\text{select}(\{p' \in P(q, a) : p' > p\}, a).$$

Again use its returned transition, leave the row undefined on reject, and use $(F_q, d(q))$ on exhausted. Every unspecified row is undefined.

All defined endmarker moves are inward. A selected backward descent that would move outward is eliminated by the boundary-leaf test. An exhausted forward row follows the source's original inward marker move. An exhausted backward row moves to its parent, and the two cases in which that parent would lie outside the tape were explicitly left undefined. Furthermore, every transition entering B_f , except its initialization row, moves $-d(f) = +1$. Consequently (B_f, L) cannot be revisited, which justifies reusing B_f as the initial state without an additional state.

To prove correctness, fix an input. Consider the source configurations whose runs under δ^* reach (f, L) . They form a finite tree directed toward the root (f, L) : each nonroot vertex has one successor, and no directed cycle can reach the root. Order the children of each vertex by source-state order. Incoming direction determinacy places all children of one vertex at the same input cell.

The prescribed machine performs depth-first traversal of this reverse tree. A backward row starts a child list; a forward row advances in that list or returns to the parent. Boundary leaves are skipped without entering them, after checking whether they are the source's initial configuration. The machine rejects immediately if it encounters (s, L) . If it finishes the tree without this encounter, it reaches (F_f, L) and accepts. Thus it rejects exactly when the source accepts. The finite tree traversal always terminates, even when the source's initial computation loops. This proves both complementation and halting with $2n$ states. $\square$

Proof of Theorem 2.1. Take $k = \lfloor n/2 \rfloor$, $\ell = \lceil n/2 \rceil$, and $K_n = L_{k, \ell}$. Corollary 6.5 gives $\text{sc}(K_n) = k + \ell = n$. The witness is direction-determinate, since all P -states are entered moving right and all Q -states moving left. Theorem 7.1 therefore supplies a halting complement with at most $2n$ states. Theorem 6.4 requires at least $2n$ states. Thus $\text{sc}(\overline{K_n}) = 2n$. $\square$

7.2. States entered in both directions

Corollary 7.2. *For an arbitrary n -state source, first delete the effective outgoing row at (f, L) . Let b be the number of states that are targets of both a left-moving transition and a right-moving transition in the resulting table, including endmarker transitions. Its complement has a halting recognizer with at most $2n + 2b$ states.*

Proof. The cases $s = f$, or $s \neq f$ with no transition entering f while moving left, have the trivial complements used above. In every other case, split each of the b mixed states q into q^- and q^+ . Keep one copy of every other state, assigned its unique incoming direction, or any direction if it has no incoming transitions. Give every copy of a source state the same outgoing rows; a row whose original target was q and whose direction was ε enters the copy of q assigned direction ε . Choose any existing copy of s as initial and the copy of f assigned direction -1 as final; denote this final copy by f^- . The resulting machine has $n + b$ states and is direction-determinate.

Projection of the cloned states to their original names reproduces each run of the original source. Since $s \neq f$, every visit to (f, L) must have arrived by a left move and therefore corresponds precisely to (f^-, L) . Thus the splitting preserves the language, including infinite rejecting runs. Theorem 7.1 gives the stated bound. $\square$

In particular, a family requiring at least cn complement states for a fixed $c > 2$ must have

$$b \geq \frac{c-2}{2} n.$$

Thus a family with only $o(n)$ mixed states has complement complexity at most $2n + o(n)$. Since always $b \leq n$, the same construction also proves $C_3(n) \leq 4n$ in the exact model of this paper. Together with Theorem 2.1, this gives

$$2n \leq C_3(n) \leq 4n \quad (n \geq 22).$$

The matching $2n$ upper bound applies to direction-determinate sources; the lower bound in Theorem 2.1 applies to unrestricted targets.

8. Comparison with GMP's acceptance convention

The exact equalities of Theorem 2.1 use left-endmarker acceptance. In GMP's definition [2, Section 2], stationary moves are allowed, there may be several final states, and a computation accepts upon reaching any final state at any tape position. Write $\text{sc}_{\text{GMP}}(L)$ for the minimum number of states under that convention. Our state Q_1 participates in ordinary backward scans, so it cannot simply be declared globally final.

This difference concerns the role of a final state, rather than the choice of the left or right endmarker. In our model the final state can do ordinary work away from the accepting marker. GMP's Lemma 3.1 [2, pp. 1176–1177] normalizes a nontrivial-language recognizer without adding states: it has one final state, that state sweeps left on every symbol other than L , its left-marker row is undefined, and there are no stationary moves. A visit to that state therefore reaches the left endmarker. Reading the normalized table with our acceptance condition preserves the language.

Conversely, an r -state automaton $A = (Q, \Sigma, \delta, s, f)$ in our model has an $(r + 1)$ -state GMP realization

$$\widehat{A} = (Q \cup \{a\}, \Sigma, \widehat{\delta}, s, \{a\}),$$

where a is fresh, $\widehat{\delta}(f, L) = (a, 0)$, all rows of a are undefined, and every other old row is unchanged. Only a is globally final. Thus for every nontrivial language L ,

$$\text{sc}(L) \leq \text{sc}_{\text{GMP}}(L) \leq \text{sc}(L) + 1.$$

The extra state does not affect the leading factor, but it matters for the exact bound. Merely applying these two inequalities gives a $2N - 2$ complement lower bound for the $N = n + 1$ state source. A fixed channel in GMP's normal form yields the stronger exact comparison below.

Lemma 8.1 (A fixed accepting channel). *Let an m -state automaton be in the normal form of GMP's Lemma 3.1: its unique final state f moves left on every symbol other than L , and its row at L is undefined. Interpret its acceptance at L only. For any group of nonempty interval diagrams obtained as in Lemma 4.3, the signed channel action has a globally fixed backward channel. Removing this channel leaves a faithful action on at most $m - 1$ points.*

Proof. For every nonempty ordinary word w , a right entry in state f sweeps left through the entire interval and exits in state f . Hence $B_w(f) = f$. In particular, for the identity diagram e , we have $f \in V = \text{Im } B_e$. For every group element h , the normal form $B_h = \beta_h B_e$ gives

$$\beta_h(f) = \beta_h(B_e(f)) = B_h(f) = f.$$

Thus the signed point (backward, f) is fixed by the homomorphic action (α_h, β_h^{-1}) . Its removal preserves faithfulness: any element fixing all remaining points also fixes the removed point, and hence is the identity. By Lemma 4.5, the remaining set has size $|U| + |V| - 1 \leq m - 1$. $\square$

Theorem 8.2 (Exact gap under GMP's convention). *For every $N \geq 23$, there is a language L over the fixed ternary alphabet Σ such that, in GMP's model,*

$$\text{sc}_{\text{GMP}}(L) = N, \quad \text{sc}_{\text{GMP}}(\overline{L}) = 2N - 1.$$

In particular, the gap asked for in [2, Section 6, p. 1186, problem (e)] exists.

Proof. Put $n = N - 1 \geq 22$ and $L = K_n$. These languages are nontrivial: $\varepsilon \in K_n$ and $\mathsf{C} \notin K_n$.

For the source upper bound, add a fresh state a to the n -state source five-tuple, make a the sole globally final state, and set $\delta(Q_1, \mathsf{L}) = (a, 0)$. Retain all other original rows and leave all rows of a undefined. No original state is globally final. This is an $(n + 1)$ -state GMP automaton for K_n . For the complement upper bound, take the $2n$ -state halting complement from Theorem 7.1. Apply the same conversion: add a fresh sole final state a , replace the accepting configuration's unused left-marker row by a stationary move into a , and keep every other original state nonfinal. This gives a $(2n + 1)$ -state GMP automaton for $\overline{K_n}$.

For either lower bound, normalize an arbitrary m -state GMP recognizer using GMP's Lemma 3.1, with no increase in states. Because its final state sweeps left, interpreting finality only at L preserves the language. Contextual separation still gives a group of actual interval diagrams mapping onto $\mathfrak{A}_k \times \mathfrak{A}_\ell$, where $k + \ell = n$. By Lemma 8.1, this group has a faithful action on at most $m - 1$ signed channels after deleting its fixed accepting channel.

For a recognizer of K_n , Corollary 5.4 requires at least $k + \ell = n$ points in this remaining action. Consequently $m \geq n + 1$.

For a recognizer of $\overline{K_n}$, suppose that $m < 2n + 1$. The remaining faithful action has fewer than $2(k + \ell)$ points, so Lemma 5.3 supplies one natural moving orbit for one semantic factor, with all other remaining channels fixed. Restoring the deleted channel adds only a fixed point. The subgroup therefore satisfies exactly the hypotheses of the query reduction in Theorem 6.4. Its four factor-and-direction cases give the same contradiction. Hence $m \geq 2n + 1$.

The matching upper and lower bounds establish the equalities $n + 1 = N$ and $2n + 1 = 2N - 1$. $\square$

For this family the GMP ratio is $(2N - 1)/N = 2 - 1/N$, tending to two. The explicit $(2N - 1)$ -state complement also shows that the same language cannot witness a $2N$ lower bound in GMP's convention. Theorems 2.1 and 8.2 give the exact counts in the two respective models.

9. Conclusion

We have exhibited a language of two-way deterministic state complexity n whose complement requires exactly $2n$ states, for every $n \geq 22$, over one fixed ternary alphabet. In GMP's convention the same

family has exact complexities N and $2N - 1$. This answers their question about the existence of a complementation gap.

The witness reaches the $2n$ upper bound for direction-determinate sources. For general sources, the bound $2n + 2b$ shows that a larger leading constant requires linearly many states that can be entered from both directions. The optimal general constant remains between two and four.

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